Ericsson PE Node JAY-PE-09 Minor QoS Congestion Event ALARM_9642 Resolution SOP

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_9642

Description

In the context of the Ericsson IP Network Node, the PE (Provider Edge) role is responsible for managing the exchange of data packets between different network domains. The QoS (Quality of Service) framework, implemented in the node's control plane, ensures that packets are prioritized and transmitted based on their relative importance. However, in scenarios where microbursts or interface tail drops occur, the QoS queue can become congested, leading to packet losses and subsequent alarms. The specific failure domain of ALARM_9642 on the JAY-PE-09 node indicates that the congestion event is related to the node's ability to manage packet transmission rates, potentially due to hardware or software limitations in the Ericsson equipment.

Pre-requisite Checks

- Check: Verify that the node's power supply is stable and within acceptable limits, Description: This is crucial to ensure that the hardware components are functioning correctly.

- Check: Confirm that all redundant links are operational and synchronized, Description: This is essential to prevent any potential issues with the node's ability to manage traffic.

- Check: Review the node's configuration and ensure that QoS settings are properly configured, Description: This is critical to prevent any potential issues with packet prioritization and transmission.

Potential Root Causes

Cause	Probability
Hardware component failure (e.g., interface card or CPU)	High
Software bug or misconfiguration in the QoS framework	Medium
Environmental factors (e.g., temperature or humidity fluctuations)	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource utilization, Expected Output: System resource utilization: CPU 50%, Memory 75%, Interfaces 60%, Description: This command provides an overview of the node's resource utilization, helping to identify potential bottlenecks.

Command: show qos policy all, Expected Output: QoS policy: Priority 1, Bandwidth 100Mbps, Buffer Size 1024, Description: This command displays the node's QoS policy configuration, allowing for analysis of packet prioritization and transmission rates.

Command: show interfaces all, Expected Output: Interface Status: UP, Speed 1000Mbps, Duplex Half, Description: This command provides an
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overview of the node's interfaces, helping to identify potential
issues with packet transmission rates.
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Resolution Steps

Step 1: Step: Clear the congestion event, Command: clear qos congestion, Description: This command clears the congestion event and resets the node's QoS framework.

Step 2: Step: Adjust QoS settings to optimize packet transmission rates, Command: configure qos policy priority 1 bandwidth 200Mbps buffer size 2048, Description: This command adjusts the node's QoS policy configuration to optimize packet transmission rates and prevent future congestion events.

Post-Check Verification

```
Command: show qos policy all, Expected Output: QoS policy: Priority 1,
Bandwidth 200Mbps, Buffer Size 2048, Description: This command
verifies that the QoS policy configuration has been successfully
updated.

Command: show interfaces all, Expected Output: Interface Status: UP,
Speed 1000Mbps, Duplex Full, Description: This command verifies that
the node's interfaces are functioning correctly and packet
transmission rates are optimized.
```

Roll-back Procedures

- Step: Revert QoS settings to original configuration, Command: configure qos policy priority 1 bandwidth 100Mbps buffer size 1024, Description: This command reverts the QoS policy configuration to its original state, in case the fix fails to resolve the issue.

- Step: Clear the congestion event, Command: clear qos congestion, Description: This command clears the congestion event and resets the node's QoS framework.